

EAS-VGGT: Evidence-Conditioned Visual Geometry for Dynamic Sensor Reconstruction

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Abstract

Feed-forward visual geometry predicts points and cameras, but it does not say which parts of a dynamic surface were observed by a range sensor, whether a surface blocks a beam, or whether that block produces a detectable return. We present EAS-VGGT, a prototype representation that adapts visual geometry into Actor-canonical continuous free/occupied/unknown evidence while separating physical state, appearance, and rigid trajectory. A late-fusion adapter improves development Brier score from .19186 to .18755 over an equal-capacity no-visual model. On a separate route-selection split, a simplex-preserving reliability discount improves Brier from .23753 to .23434 and NLL from .95972 to .94380 on 239 candidates, while leaving the inherited surface essentially unchanged. A controlled ordered-return experiment shows why blocking and detection must be separate: its NLL is .86215 versus 1.05022 for a single-hazard model. We also verify an appearance branch that gains .545 dB without updating physical geometry and SE(3) composition with numerical errors below 7×10^{-13} . A frozen source confirmation is inconclusive because its preselected views contain zero observable Actor candidates; we report that support failure rather than score an empty set. The results support the representation mechanisms and expose geometry/generalization as the remaining work for a competitive system.

1. Introduction

Large visual geometry models can reconstruct dense point maps from a few RGB views, yet a world simulator must answer a different question: what would a calibrated sensor measure after dynamic objects move? A visually plausible point does not establish free-space support, blocking probability, detectability, or a no-return outcome. Treating RGB opacity as LiDAR opacity also couples appearance edits to physical sensing.

We study this gap without replacing the existing dynamic world model with a generic completion network. EAS-VGGT starts from the positive components of our prior system: Actor-canonical surface anchors, categorical return ev-

idence, independent appearance capacity, and rigid owner trajectories. It adds visual features as a measurement expert, maintains continuous free/occupied/unknown evidence, and composes scene events in ray order. Physical state and appearance share ownership and motion but have separate trainable parameters.

The experiments reveal a useful boundary. Direct early fusion overfits, and visual features from π^3 do not transfer in this implementation. Late fusion with VGGT does improve held-out evidence calibration, although its surface displacement does not improve geometry. Raw visual fusion is slightly harmful on route selection, while a fixed convex reliability discount recovers a small gain in both proper scores. Thus the defensible contribution is observation-aware evidence adaptation, not global dense completion.

Our concrete contributions are:

- a ray-preserving EAS data and I/O contract that keeps build measurements, held-out targets, Actor ownership, and split roles separate;
- a continuous evidence adapter and a normalized ordered-return measure that distinguishes blocking from detection and valid no-return;
- a physical/appearance ownership bridge and a shared SE(3) trajectory action;
- an executed comparison across VGGT, π^3 , MapAnything, simple evidence baselines, and a neural-LiDAR capability reference, including negative results and an explicit insufficient-support source outcome.

2. Related Work

Feed-forward visual geometry. VGGT jointly predicts cameras, depth, point maps, and tracks from multiple views [4]. π^3 removes a distinguished reference view through permutation-equivariant geometry prediction [5]. MapAnything accepts optional intrinsics, poses, depth, and metric-scale prompts in a unified model [1]. These are strong geometry priors, but none directly defines sensor free/occupied/unknown evidence or an ordered first-return distribution.

Robust multimodal fusion. Fusion may fail when one modality is missing or degraded. Work on missing-modality

transformers [2] and multi-modal expert routing [3] motivates keeping visual and measurement experts separately usable. Our reliability discount is a small post-hoc instance of this principle; it is selected only on route data and does not constitute a new general calibration method.

Neural LiDAR and dynamic rendering. LiDAR4D models geometry, intensity, ray drop, and dynamic scenes in a neural field [6]. Gaussian-splatting systems provide high-quality RGB appearance but their opacity is a rendering quantity. EAS instead keeps a sensor evidence measure, an appearance state, and an Actor trajectory as distinct objects.

3. Task Definition

For Actor i , build observations provide calibrated LiDAR rays and RGB views, an identity, and known poses $T_i(t) \in \text{SE}(3)$. Held-out points, ray outcomes, and target RGB are labels and never enter the forward pass. The representation is

$$S_i = (P_i, E_i, A_i, T_i),$$

where P_i is an Actor-canonical physical surface, E_i is continuous measurement evidence, and A_i is appearance. For a world ray $q = (o, d, r_{\min}, r_{\max})$, the model predicts a normalized distribution over ordered surface returns and no-return.

The intended task is sparse-build dynamic reconstruction and sensor replay under a given trajectory. It is not future trajectory prediction. Invalid or unfired queries, valid no-returns, unsupported geometry, background events, and Actor events remain separate. Geometry is evaluated independently from the return reader so a learned decoder cannot hide a weak surface.

4. EAS-VGGT

4.1. Continuous Evidence Adaptation

Build LiDAR is compiled into Actor-canonical surface candidates with continuous free/occupied/unknown masses and opportunity counts. A frozen visual backbone supplies point-map geometry and local features. The successful adapter first encodes the measurement-only state and then applies a visual residual to its evidence logits and surface displacement. This late-fusion form preserves a usable fallback expert; an equal-capacity model removes visual inputs.

Given visual and fallback distributions $p_v, p_0 \in \Delta^2$, route selection chooses a scalar reliability α and returns

$$p_\alpha = p_0 + \alpha(p_v - p_0).$$

The convex combination is normalized and uses the same operation for surface residuals. It is a post-hoc reliability

control, not a claim that one global α handles every domain. When the visual cohort has no observable candidates, the metric is undefined and the system reports insufficient support.

4.2. Ordered Return Measure

After Actor placement, all candidate surface events are sorted by ray depth. Event k has blocking probability u_k and conditional detection probability v_k . With $T_k = \prod_{\ell < k} (1 - u_\ell)$,

$$q_k = T_k u_k v_k, \quad q_\emptyset = T_{K+1} + \sum_k T_k u_k (1 - v_k).$$

Thus $\sum_k q_k + q_\emptyset = 1$. A front surface may block the background yet fail to produce a return, which a single hazard probability cannot express. Primitive mass is conserved under splitting so the reader does not change merely because a surface is resampled more densely.

4.3. Appearance and Rigid Motion

Appearance primitives keep a physical parent and Actor owner, but RGB loss updates only A_i . The forward link uses $A_i = H_{\text{app}}(\text{stopgrad}(P_i), f_{\text{rgb}})$, retaining correspondence while blocking RGB gradients from physical state and trajectory. Both branches use the same $T_i(t)$. Canonical queries obey $Q_{i,t}(x) = Q_i(T_i(t)^{-1}x)$; under a common transform g , $Q_{gT}(gx) = Q_T(x)$ and ordered ray distances are preserved.

5. Experiments

Data and protocol. nuScenes logs are separated into fit/development, route-selection, and source roles. The route corpus contains 31 scenes and 310 Actors (142 hazardous), with 816,442 build points and 284,030 target rays. Its visual evaluation covers 35 Actors and 239 surface candidates. The source I/O recovery materializes all 677 required keyframe scans and 80 Actors from three frozen logs. Model selection never uses source labels.

Foundation diagnostics. Official VGGT-1B and π^3 checkpoints run through one shared camera/point-map interface. Their first raw LiDAR-alignment diagnostic gives median visible-surface residuals of 74.67 m and 3.41 m, respectively, after the then-current similarity alignment. These values diagnose adapter/alignment risk rather than benchmark the backbones. MapAnything is then run through its native calibrated pose and intrinsic prompts on a route window; it reaches 0.799 m median surface residual and 4.76% hit at 0.2 m. It is a stronger metric reference, but still does not supply range-sensor evidence directly.

Table 1. Evidence prediction on development. All learned rows use comparable small adapters; lower is better. “Early” denotes direct joint fusion.

VGGT arm	F/O/U NLL	Brier
Build evidence input	1.24246	.51723
Scalar early fusion	1.51519	.41679
F/O/U early fusion	.95373	.27455
Late, no visual	.79168	.19186
Late EAS-VGGT	.79123	.18755

Table 2. Frozen route-selection evidence scores on 239 candidates.

Arm	NLL↓	Brier↓
Build evidence input	1.31308	.47670
Late, no visual	.95972	.23753
Raw visual residual	.95848	.24477
Reliability $\alpha = .35$	.94380	.23434

Table 3. Controlled ordered-return intervention (16,384 held-out rays).

Reader	NLL↓	Brier↓	BG MAE↓
Single hazard	1.05022	.11969	.22722
Block/detect	.86215	.000005	.000423

Late fusion is the transferable development result. Table 1 shows that the original joint adapter overfits relative to the late-fusion decomposition. A near-capacity scalar head also generalizes poorly. The VGGT visual residual improves Brier by .00431 over its no-visual counterpart while surface displacement RMSE is effectively unchanged (.07530 versus .07549 m). Replacing VGGT with π^3 worsens Brier from .19264 to .24066, so cross-backbone universality is not supported.

Reliability discount. Raw visual fusion slightly improves route NLL but harms Brier. Selecting $\alpha = .35$ on route data gives the best Brier among the fixed grid and improves both proper scores over the no-visual expert (Table 2); two of three logs improve in Brier. Mean surface F-score remains .66756 and symmetric CD remains approximately .21031 m for every learned arm. This is evidence calibration, not geometry repair.

Composition mechanisms. The separated blocking/detection reader remains better after moving the front Actor behind the background (NLL .67563 versus .93539), and mass splitting changes the output by zero numerically. The appearance bridge assigns all 309 physical parents to 3,090 visual primitives. Across six held-out views it improves pooled Actor PSNR from 17.186 to 17.730

dB; physical geometry and trajectory receive no RGB update. Its 25.35 dB StreetGS reference also records the remaining appearance gap. On 11 Actors and 40 poses, maximum SE(3) commutation, Actor-frame recovery, and pairwise-distance errors are 6.82×10^{-13} , 1.48×10^{-13} , and 2.82×10^{-13} m; the largest observed translation span is 5.24 m.

Source support and neural-LiDAR boundary. The frozen source visual set contains 12 windows and eight matched Actor poses, but zero observable surface candidates and zero camera observations. Proper evidence scores are therefore undefined. A target-free audit records `inconclusive_insufficient_visual_support`; the route result is not promoted to an independent-generalization claim. Separately, an official LiDAR4D capability run on KITTI-360 completes range, ray-drop, intensity, and scan export (ray-drop F1 .9553, depth RMSE 2.947 m). Its native protocol is not numerically comparable to the Actor task.

6. Limitations

The positive real-data gain is small and appears in evidence calibration, while the physical surface is unchanged. π^3 does not reproduce the VGGT gain, and the MapAnything run is a diagnostic rather than an adapted EAS baseline. The ordered-return result uses a controlled analytic generator; real valid no-return and background composition still require a matched full-scene dataset. Appearance improves only one Actor sequence and remains well below the StreetGS reference. Exact SE(3) identities validate the representation, not pose estimation or future-motion prediction.

Most consequentially, the frozen source scenes have no visual candidate support. A future independent protocol must predeclare a target-free eligibility threshold for camera/Actor overlap and evaluate the fallback expert when a modality is absent. It must then freeze that rule before opening quality. The current source data cannot be reused to claim independent confirmation. No occupancy, complete waveform, collision, planning, closed-loop, or safety claim follows from these experiments.

7. Conclusion

EAS-VGGT reorganizes dynamic world simulation around owned physical state, continuous measurement evidence, independent appearance, and a shared rigid trajectory. The prototype establishes four concrete mechanisms: late visual evidence adaptation, ordered blocking/detection composition, gradient-isolated appearance with parent correspondence, and exact SE(3) action. The real-data evidence gain is modest and does not improve geometry, while source confirmation lacks support. The next publishable step is conse-

quently clear: learn localized physical surface events on a support-eligible independent cohort and compare them under a complete first-return protocol.

References

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